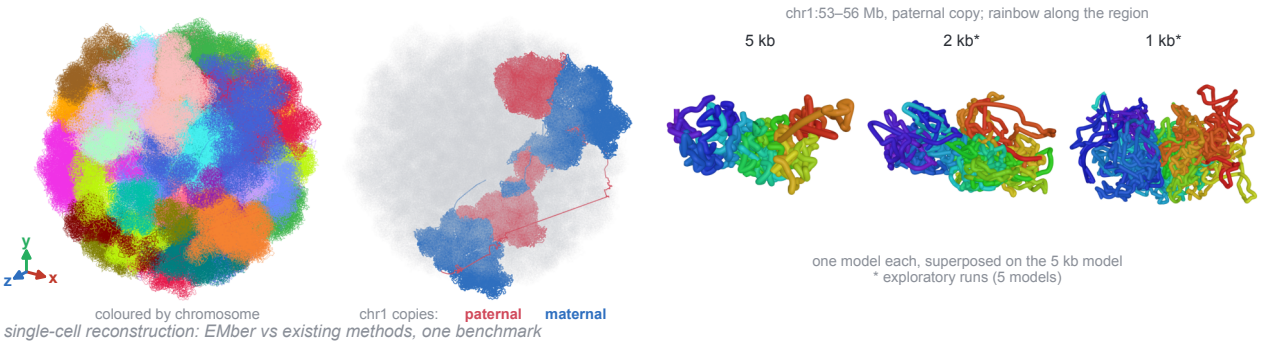
single-cell reconstruction at high resolution (EMber, homolog-resolved)

a **EMber (U-Chrom): a diploid human genome at 5 kb**

GM12878 scMicro-C (Wu 2025), deepest cell: 3.01 M contacts, 1.16 M particles (2 × 23 chains); 10 models in 4.6 min on one H100 (2.8 GB)

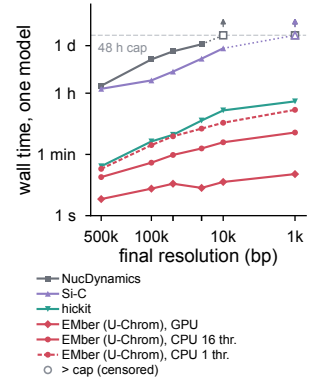
one model shown; low run-to-run agreement on this data (seeds: r = 0.47 per copy chain; the authors' hickit runs: 0.98); Supplementary Fig. S8



single-cell reconstruction: EMber vs existing methods, one benchmark

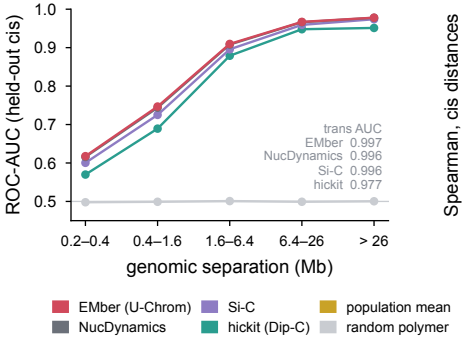
b **Time for one model**

Cell 1; CPU: EPYC 9754, GPU: H100



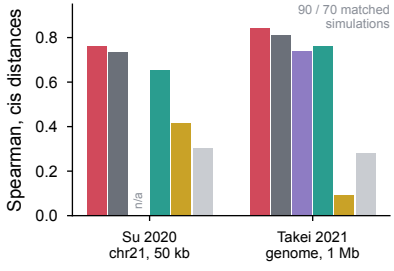
c **Held-out contacts**

8 cells × 3 folds; matched-separation negatives



d **Imaging truth (simulated Hi-C)**

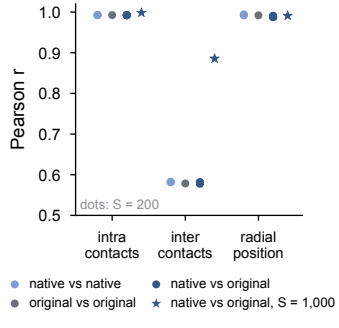
median model vs imaged distances



population reconstruction (IGM)

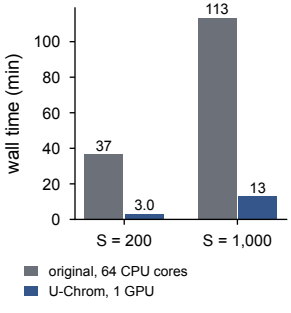
e **IGM: native vs original**

WTC11 2 Mb; population agreement



f **IGM speed**

same input and schedule



g **How many structures?**

WTC11 2 Mb, 3 seeds; grey: minutes on 1 GPU

